



## Research paper

## Applying artificial neural network to zero-crossing wave parameters for the wave spectrum

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## ABSTRACT

This paper presents a neural network model to estimate wave spectra from wave parameters by the zero-crossing analysis, maximum, significant, and average wave heights and their corresponding wave periods. The present model was trained with the wave data in the JES (Japan/East Sea), and then applied to wave measurements in the Yellow Sea and the South Sea for its feasibility at other seas. According to the present study, the predicted wave spectra well agree with the measurements, even the wave data not in model training. This comparison shows that the accuracy of the present model is comparable to that of JONSWAP wave spectrum. Furthermore, the present model was employed to the wave hindcasting to see its applicability, while JONSWAP wave spectrum also being applied. The significant wave heights from both numerical simulations well agree with the measured ones. However, the average RMSE of significant wave period with the present neural network was 7.08 % smaller than the other, and its Pearson's correlation coefficient is 6.33 % larger on average. From this, it can be seen that the present model was well applied to the wave hindcasting without considering the local wave measurements.

## 1. Introduction

The wave parameters, wave heights and periods, are obtained from the zero-crossing analysis and spectral analysis, whose details are given in [Dean and Dalrymple \(1991\)](#). The wave parameters from both methods are different to each other, but several wave parameters by the zero-crossing analysis can be also obtained from the spectral analysis ([Chun and Suh, 2018, 2019](#)). Besides, the effect of the spectral shapes on the coastal environment has been reported in several studies. According to [Goda \(2008\)](#), the wave setup increases along with the peak enhancement factor of JONWAP wave spectrum, and this parameter influences on the location of maximum longshore currents velocity as well. [Bakali et al. \(2022\)](#) have shown that the spectral shapes give a significant effect on the wave overtopping and pressure in the front of the breakwaters.

Most of wave measurements do not provide wave spectra, in spite of their importance. Accordingly, wave spectrum models such as JONSWAP (Joint North Sea Wave Observation) and TMA (Texel-Marsen-Arsloe) are used to describe wave spectra. Though obtained from the

field study ([Hasselmann et al., 1973](#); [Hughes, 1984](#)), the wave spectrum models have been applied to predict wave spectra well. Meanwhile, several researchers improved them, adjusting their parameters. [Mansard and Funke \(1990\)](#) provided the improved parameter estimation of JONSWAP wave spectrum model, where the peak frequency, the peak enhancement factor, and the zeroth order of wave spectrum were estimated considering the wave spectral shape. [Hwang et al. \(2021\)](#) proposed a formula for the peak enhancement factor of JONSWAP wave spectrum, which was written in terms of significant wave height, peak frequency, and spectral peak density. [Mazzaretto et al. \(2022\)](#) suggested the peak enhancement factor over the global coastal region, which enhanced the suitability of the JONSWAP wave spectrum. These studies have improved the accuracy of the wave spectrum models, but all wave spectra cannot be expressed by these wave spectrum models.

On the other hand, [Naithani and Deo \(2005\)](#) have applied the ANN (Artificial Neural Network) to predict wave spectra. In their model, four wave parameters from the spectral analysis, significant wave height, peak wave period, peakedness parameter, and spectral width parameter were used to produce wave spectra, where the number of spectral

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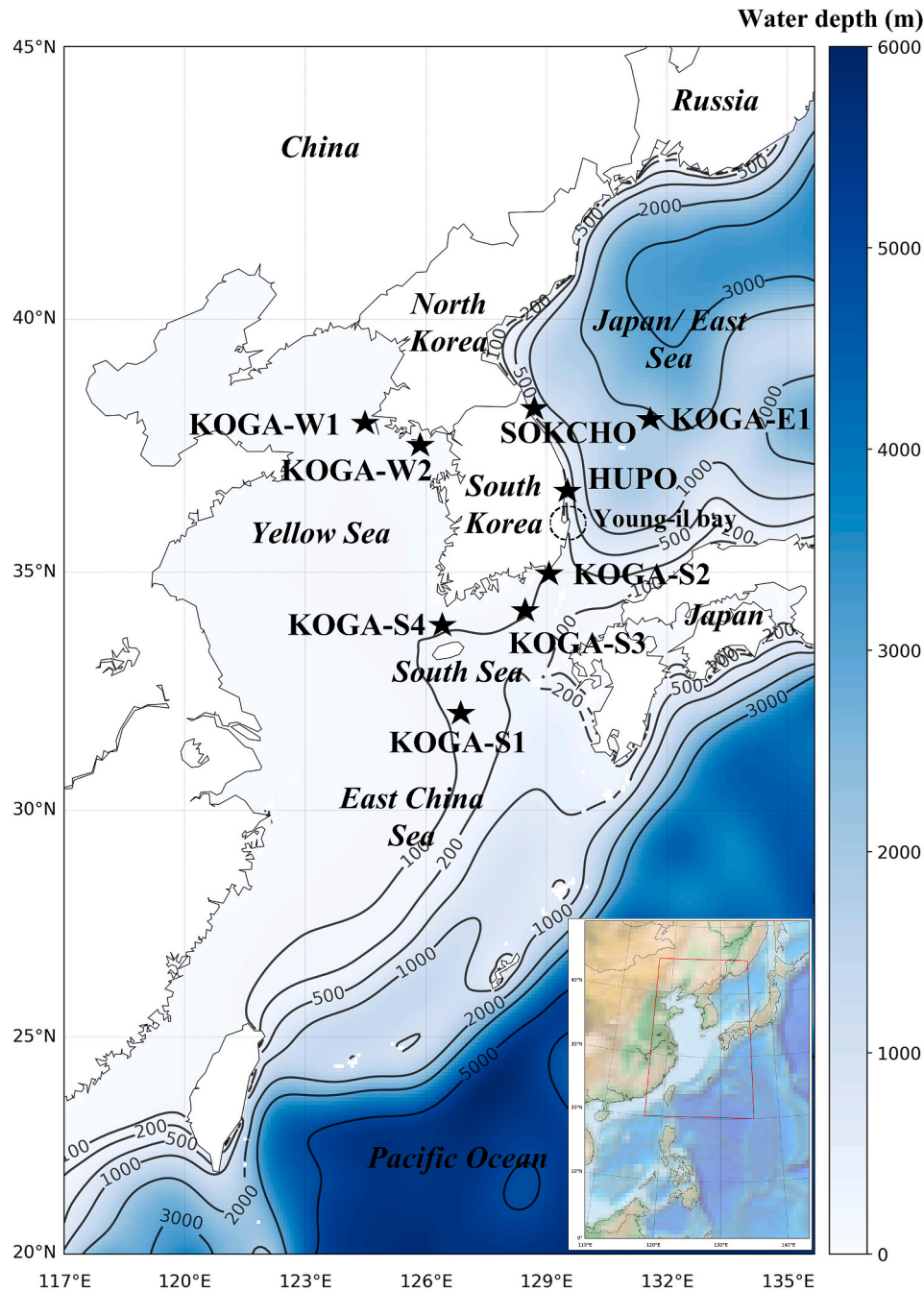


Fig. 1. Location of the wave measurements.

frequency was 24. Among the various methods of ANN, the MLP (Multi-Layer Perceptron) was chosen to generate wave spectra at Goa, India and Cape, Hatteras, U.S., and the error between the predicted and measured wave spectra was measured as MSE (Mean Square Error). According to their study, there were good agreements between the measured and predicted wave spectra, particularly, well reproducing peak frequency and spectral peak density. However, spectral energy densities in low and high frequencies were overestimated, because their contribution to MSE is too small to update the coefficients of network model accurately. This might result from that most of the spectral energy density is distributed around the peak frequency. Moreover, the network model was not validated against the data not included in model training.

Next, [Namecar and Deo \(2006\)](#) have also provided the ANN for wave spectra, where two parameters, significant wave height and mean wave period were used. Although the number of wave parameters of [Namecar](#)

and Deo's (2006) model was smaller than that of [Naithani and Deo \(2005\)](#) and the spectral shape parameters were not included in the model, the spectral shapes by the network model well agreed with the measured ones. Especially, spectral energy densities at high and low frequencies were not overestimated unlike [Naithani and Deo \(2005\)](#), resulting from that the error of the network model was estimated by the correlation coefficient, not MSE. Then the estimated error was backward propagated to update the weights and biases in the hidden layers more accurately. Meanwhile, the training data were randomly shuffled, considering 721 data. This is helpful to the generality of the network model, but this was not also validated against the wave data not included in model training.

Differently with these studies, [Hwang et al. \(2021\)](#) have obtained wave spectra from the probability distribution of individual wave heights and periods after the zero-crossing analysis, which is called

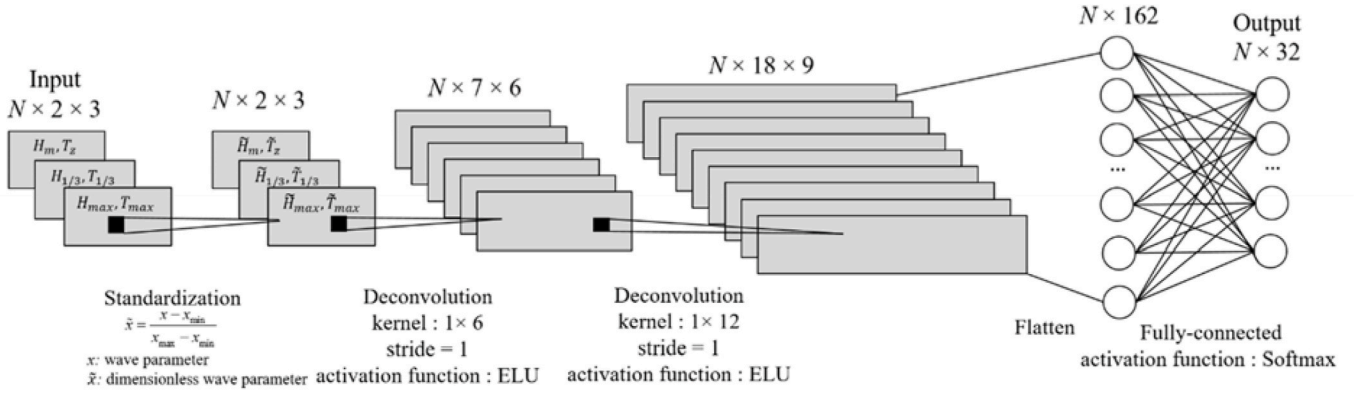


Fig. 2. Model architecture for predicting wave spectra.

ZUCT (Zero Up-Crossing Transform). Being based on the zero-crossing analysis, the method is advantageous in representing the wave spectra of the nonlinear wave. However, it is somewhat inefficient, requiring individual waves from the zero-crossing analysis. Furthermore, the method cannot provide wave spectra unless the surface elevation is absent. For this reason, it was suggested in the present study that ANN model was applied to obtain wave spectra from the wave parameters by the zero-crossing analysis, maximum, significant, and average waves.

In the present study, DeCNN (Deconvolutional Neural Network) was employed to obtain wave spectra, which expanded maximum, significant, and average waves into wave spectra. DeCNN has been widely applied to the field of the computer vision (Charte et al., 2018; Zeiler et al., 2010). Furthermore, the KL (Kullback-Leibler)-divergence was used as the loss function of the present neural network, which measures the difference of two distributions (Charniak, 2018). The present model was implemented using the wave parameters from the zero-crossing analysis, not the spectral analysis. These are different from the previous neural network models on the wave spectrum. On the other hand, the present model was built from the wave data in Hupo, KOGA-E1, and KOGA-S2 (Fig. 1), and then applied to the wave measurements at other seas, the Yellow Sea and the South Sea to see its feasibility. According to the present study, there were good agreements between the predicted and measured wave spectra, not considering the local wave characteristics. Additionally, its applicability was seen by applying it to the wave hindcasting in Young-il bay. According to the present numerical results, the significant wave period was more accurately predicted, when the incident wave by the present neural network model was applied to the numerical experiment.

## 2. Methods

### 2.1. Neural network model

The zero-crossing analysis provides maximum wave height,  $H_{max}$ , maximum wave period,  $T_{max}$ , significant wave height,  $H_{1/3}$ , significant wave period,  $T_{1/3}$ , average wave height,  $H_m$ , and mean wave period,  $T_z$ . These wave parameters were expanded to predict wave spectra in the present neural network model. For the neural network, the wave parameters were standardized with respect to the maximum and minimum of each parameter. Here, the standardization is given as  $\tilde{x} = (x - x_{min}) / (x_{max} - x_{min})$ , where  $\tilde{x}$ ,  $x_{max}$ , and  $x_{min}$  are a dimensionless wave parameter, a maximum wave parameter, and a minimum wave parameter, respectively, for a wave parameter,  $x$ . According to Fig. 2, the present neural network was composed of two deconvolution layers and one fully-connected layer. Here, the one-dimensional deconvolution kernel was used in the deconvolution layer, whose operation is similar to that of convolution kernel. In Fig. 2,  $N$  denotes the number of wave records.

The network model was trained, repeating the steps of forward and backward propagations. During a forward step, the intermediate variables of  $(l + 1)$ -th layer,  $z^{l+1, f_1}$  from the deconvolution and fully-connected layers are calculated by Eqs. (1) and (2), respectively.

$$z_i^{l+1, f_1} = \sum_{f_2} (w^{l, f_1, f_2} \otimes a_i^{l, f_2}) + b^{l, f_1}, l = 1, 2 \quad (1)$$

$$z_i^{l+1, f_1} = \sum_{f_2} (w^{l, f_1, f_2} \times a_i^{l, f_2}) + b^{l, f_1}, l = 3 \quad (2)$$

where  $i$ ,  $f_1$ , and  $f_2$  represent the indices of wave record, deconvolution kernel, and channel, respectively. Applying a number of deconvolution kernels or weight matrices to each layer of the network model, outputs from these deconvolution operations or matrix multiplication put into channels of internal variables, which are presented as the third dimension in Fig. 2. Thus third dimensions of nodes of the first and second layers are 6 and 9, respectively, which are corresponding to the numbers of deconvolution kernels of the layers. In addition to this, the operator,  $\otimes$ , indicates the deconvolutional operation.  $w^{l, f_1, f_2}$  and  $W^{l, f_1, f_2}$  are the deconvolution kernel and weight matrix, respectively.  $b^{l, f_1}$  is the intercept. Further,  $a^{l, f_1}$  is obtained by putting  $z^{l, f_1}$  into the activation function. In the present network model, two activation functions, ELU (Exponential Linear Unit) and softmax, were employed, whose details are given by Goodfellow et al. (2016). Here, softmax was applied to the last layer of the present neural network, as shown in Fig. 2. Owing to this, each spectral energy density was normalized as follows

$$\hat{S}_{ij} = S_{ij} / \sum_{j=1}^{N_f} S_{ij} \quad (3)$$

where  $j$  denotes the index of element in the spectral energy density and  $N_f$  is the number of spectral energy components.  $S_{ij}$  is the  $j$ -th element of the  $i$ -th measured wave spectrum. Here,  $S_{ij}$  was given by applying ZUCT to the wave measurements, because the present study was extended from Hwang et al. (2021).  $\hat{S}_{ij}$ , the normalized spectral energy density, was incorporated to estimate the error, which is as follows

$$\epsilon^2 = \frac{1}{N} \sum_{i=1}^N \sum_{j=1}^{N_f} \hat{S}_{ij} \log(\hat{S}_{ij} / a_{ij}^{l+1}) + \frac{\lambda_r}{2} \left( \sum_r \sum_c (w_{r,c}^{l, f_1, f_2})^2 + \sum_c (w_c^{l, f_1, f_2})^2 \right) \quad (4)$$

where  $a_{ij}^{l+1}$  is then predicted values at the output layer of the neural network. When predicting wave spectra,  $a_{ij}^{l+1}$  is scaled with the corresponding significant wave height, transformed into  $S_{ij}$ .  $r$  and  $c$  are the row and column indices of deconvolution kernel or weight matrix,

**Table 1**  
Description of the wave measurements.

| Location | Longitude (°) | Latitude (°) | Measurement Period          | Measuring devices                  | Number of data | Duration time | Sampling rate |
|----------|---------------|--------------|-----------------------------|------------------------------------|----------------|---------------|---------------|
| Hupo     | 129.4210      | 36.6805      | Nov/12/2011~ Apr/2/2014     | bottom-mounted acoustic wave gauge | 31,875         | 640 s         | 2 Hz          |
| KOGA-E1  | 131.5573      | 38.0301      | Nov/22/012 ~ Oct/31/2014    | waverider buoy                     | 100,752        | 640 s         | 2 Hz          |
| KOGA-S1  | 126.9658      | 32.0903      | Aug/11/2011 ~ Aug/31/2012   | waverider buoy                     | 102,938        | 640 s         | 2 Hz          |
|          |               |              | Nov/18, 2013 ~ Oct/31, 2013 |                                    |                |               |               |
| KOGA-S2  | 129.1380      | 34.9355      | Dec/27/2012 ~ Nov/6/2014    | waverider buoy                     | 106,959        | 640 s         | 2 Hz          |
| KOGA-S3  | 128.4188      | 34.2225      | Mar/27/2013 ~ Oct/31/2014   | waverider buoy                     | 79,523         | 640 s         | 2 Hz          |
| KOGA-S4  | 126.4922      | 33.9117      | Sept/1/2011~Aug/29/2012     | waverider buoy                     | 153,788        | 640 s         | 2 Hz          |
| KOGA-W1  | 124.5338      | 37.9916      | Dec/1/2011 ~ Apr/30/2012    | waverider buoy                     | 133,207        | 640 s         | 2 Hz          |
|          |               |              | Dec/1/2012 ~ Oct/31/2014    |                                    |                |               |               |
| KOGA-W2  | 125.7322      | 37.5562      | Nov/8/2012 ~ Sep/30/2014    | waverider buoy                     | 94,238         | 640 s         | 2 Hz          |
| Sokcho   | 128.6094      | 38.2983      | Oct/24/2012 ~ Jan/22/2014   | bottom-mounted acoustic wave gauge | 28,874         | 640 s         | 2 Hz          |

respectively. The first term of Eq. (4) is the error by the KL-divergence and the second term, a relaxation term, was introduced to reduce the overfitting (Ng, 2011).  $\lambda_r$  in the second term of Eq. (4) indicates the relaxation parameter, which is given as 0.05 in the present study. In Eq. (4),  $\epsilon^2$  is an error, which is also called the loss of the neural network (Charniak, 2018). Then the error estimated by Eq. (4) is propagated backward to update the deconvolution kernel, weight matrix, and bias, which are as follows:

$$w_c^{l f_1 f_2} = w_c^{l f_1 f_2} - \alpha_1 \frac{\partial \epsilon^2}{\partial w_c^{l f_1 f_2}} + \lambda_r w_c^{l f_1 f_2} \quad (5)$$

$$w_{r,c}^{l f_1 f_2} = w_{r,c}^{l f_1 f_2} - \alpha_2 \frac{\partial \epsilon^2}{\partial w_{r,c}^{l f_1 f_2}} + \lambda_r w_{r,c}^{l f_1 f_2} \quad (6)$$

$$b_j^{l f_1} = b_j^{l f_1} - \alpha_3 \frac{\partial \epsilon^2}{\partial b_j^{l f_1}} \quad (7)$$

where  $\alpha_1$ ,  $\alpha_2$ , and  $\alpha_3$  are the learning rate coefficients to be determined by ADAM (Adaptive moment estimation) method (Kingma and Ba, 2015). The correction terms in Eqs. (5)–(7) are like the followings.

$$\frac{\partial \epsilon^2}{\partial w_c^{l f_1 f_2}} = \sum_{k=1}^{N_l} \sum_{i=1}^N \delta_{i,s \times k + c - 2}^{l+1 f_1} a_{i,k}^{l f_2}, \quad \frac{\partial \epsilon^2}{\partial w_{r,c}^{l f_1 f_2}} = \sum_{i=1}^N \delta_{i,r}^{l+1 f_1} a_{i,c}^{l f_2}, \quad \frac{\partial \epsilon^2}{\partial b_j^{l f_1}} = \sum_{i=1}^N \delta_{i,j}^{l+1 f_1} \quad (8)$$

where,  $s$  is the number of the stride in the deconvolution layer.  $N_l$  is the number of elements of internal variables in the  $l$ -th layer. The term in Eq. (8),  $\delta_{i,j}^{l+1 f_1}$  is the error term of  $(l+1)$ -th layer, which is defined as  $\delta_{i,j}^{l+1 f_1} = \partial \epsilon^2 / \partial z_{i,j}^{l+1 f_1}$ . When the deconvolution or fully-connected layers, the error term of  $l$ -th layer,  $\delta_{i,j}^{l f_2}$  is calculated as Eq. (9) or Eq. (10).

$$\delta_{i,j}^{l f_2} = \frac{\partial \epsilon^2}{\partial z_{i,j}^{l f_2}} = \left( \sum_{k=1}^{N_k} \frac{\partial \epsilon^2}{\partial z_{i,s \times j + k - 2}^{l+1 f_1}} w_{k,j}^{l f_1 f_2} \right) a' \left( z_{i,j}^{l f_2} \right) = \left( \sum_{k=1}^{N_k} \delta_{i,s \times j + k - 2}^{l+1 f_1} w_{k,j}^{l f_1 f_2} \right) a' \left( z_{i,j}^{l f_2} \right) \quad (9)$$

**Table 2**  
Statistical summary of wave measurements.

| Area       | Wave measurements | Water depth (m) | $H_{max}$ (m) | $T_{max}$ (s) | $H_{1/3}$ (m) | $T_{1/3}$ (s) | $H_m$ (m) | $T_z$ (s) |
|------------|-------------------|-----------------|---------------|---------------|---------------|---------------|-----------|-----------|
| JES        | Hupo              | 17              | 1.39          | 6.31          | 0.86          | 6.17          | 0.54      | 4.82      |
|            | KOGA-E1           | 1306            | 2.14          | 6.06          | 1.38          | 6.01          | 0.89      | 4.91      |
|            | Sokcho            | 25              | 1.27          | 5.82          | 0.77          | 5.62          | 0.47      | 4.25      |
|            | Average           | –               | 1.60          | 6.06          | 1.00          | 5.93          | 0.63      | 4.66      |
| South Sea  | KOGA-S1           | 114             | 2.51          | 6.65          | 1.62          | 6.51          | 1.05      | 5.24      |
|            | KOGA-S2           | 94              | 1.76          | 6.06          | 1.13          | 5.89          | 0.73      | 4.72      |
|            | KOGA-S3           | 77              | 1.71          | 5.81          | 1.09          | 5.63          | 0.71      | 4.56      |
|            | KOGA-S4           | 49              | 1.89          | 5.68          | 1.21          | 5.58          | 0.78      | 4.61      |
| Yellow Sea | Average           | –               | 1.79          | 5.85          | 1.14          | 5.70          | 0.74      | 4.63      |
|            | KOGA-W1           | 63              | 1.33          | 5.11          | 0.85          | 5.06          | 0.55      | 4.24      |
|            | KOGA-W2           | 15              | 0.77          | 4.51          | 0.48          | 4.42          | 0.32      | 3.70      |
|            | Average           | –               | 1.29          | 5.16          | 0.83          | 5.06          | 0.53      | 4.19      |

$$\delta_{i,j}^{l f_2} = \frac{\partial \epsilon^2}{\partial z_{i,j}^{l f_2}} = \left( \sum_{k=1}^{N_k} \frac{\partial \epsilon^2}{\partial z_{i,k}^{l+1 f_1}} w_{k,j}^{l f_1 f_2} \right) a' \left( z_{i,j}^{l f_2} \right) = \left( \sum_{k=1}^{N_k} \delta_{i,k}^{l+1 f_1} w_{k,j}^{l f_1 f_2} \right) a' \left( z_{i,j}^{l f_2} \right) \quad (10)$$

where,  $N_k$  is the size of deconvolution kernel.  $a' \left( z_{i,j}^{l f_2} \right)$  is an activation function differentiated with respect to  $z_{i,j}^{l f_2}$ . The weights updated throughout Eqs. (5)–(7) are involved in the next forward propagation. After that, the error is estimated again. Like this, the calculations are repeated until when the error satisfies the criteria. Finally, these calculations provide wave spectra. Here, the number of elements of the wave spectra by the present model is 32 for the model efficiency. The corresponding maximum frequency is 1 Hz.

## 2.2. Building the network

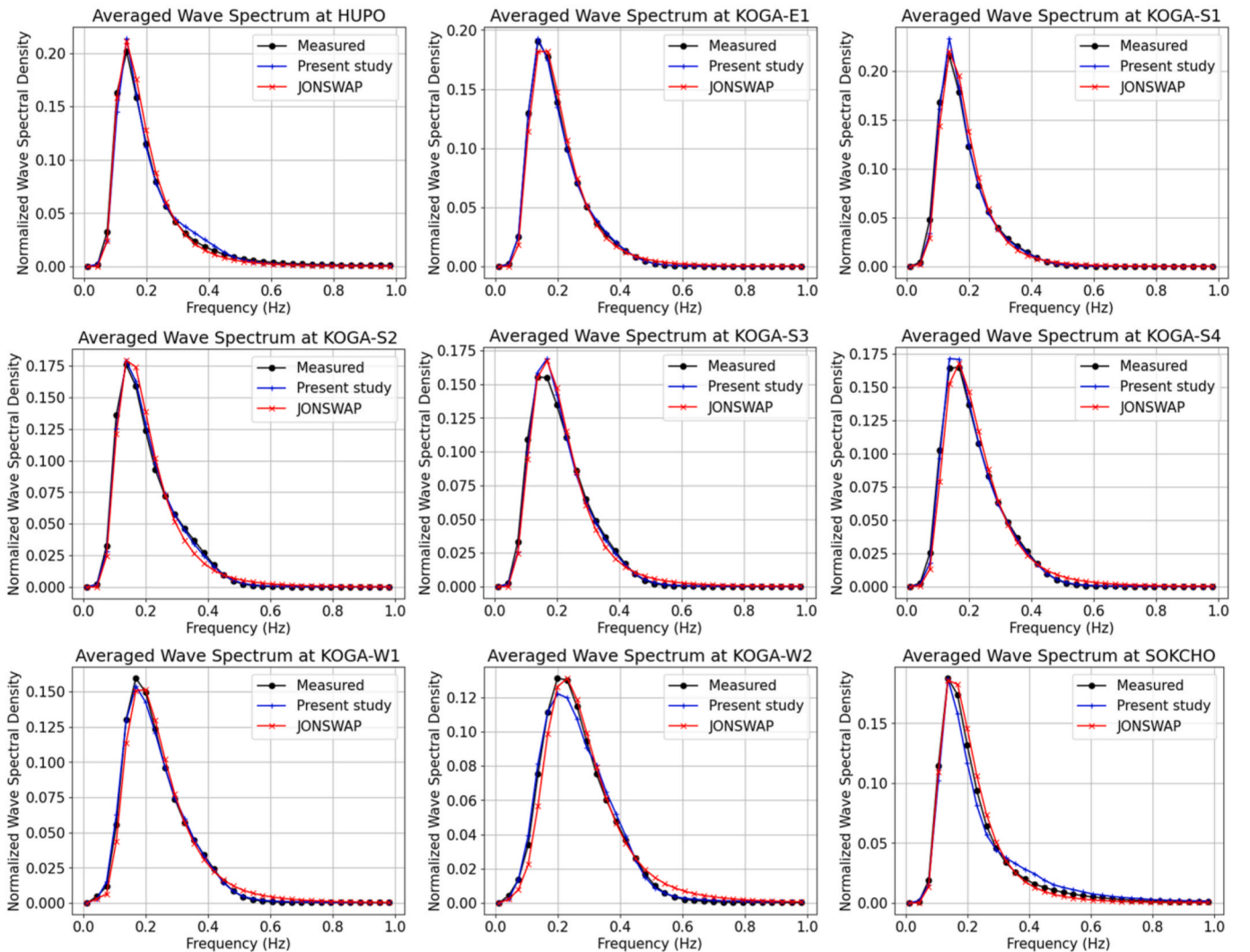
The wave measurements around the Korean peninsula were considered in the present study (Fig. 1), whose details are described in Table 1. The statistical summary of wave measurements is given in Table 2, which shows the average wave period in the JES is the largest, because its fetch length is larger than those of other seas. However, the average wave height in the JES is not larger than that of the South Sea, because the water depths at Hupo and Sokcho are shallow (Table 2) and the wave measurements in the South Sea, KOGA-S1 and KOGA-S3, are opened to the Pacific Ocean (Fig. 1). Although KOGA-S4 is located in the strait between an island and the mainland, its average wave height is not much smaller than that of KOGA-S1, because the waves at KOGA-S1 are transformed to this point. Further, several typhoons affected to this point during the measuring period. On the other hand, the average wave height and period in the Yellow Sea are the smallest owing to its small area. Due to this, wave characteristics of KOGA-W1 and KOGA-W2 are quite different from that of the JES. The difference of wave characteristics between the JES and the Yellow Sea was described in Chun and Ahn (2018).

Of the wave data, the wave measurements in the JES, specifically, Hupo, KOGA-E1, and KOGA-S2 were used to build the network model,

**Table 3**

Accuracy of wave spectra predicted by the present neural network and JONSWAP.

| Location | Present model          |                                   | JONSWAP                |                                   |                         |
|----------|------------------------|-----------------------------------|------------------------|-----------------------------------|-------------------------|
|          | error by KL-divergence | Pearson's correlation coefficient | error by KL-divergence | Pearson's correlation coefficient | Peak enhancement factor |
| Hupo     | 0.09                   | 0.93                              | 0.52                   | 0.95                              | 1.15                    |
| KOGA-E1  | 0.07                   | 0.95                              | 0.50                   | 0.95                              | 1.23                    |
| KOGA-S1  | 0.07                   | 0.94                              | 0.84                   | 0.93                              | 1.15                    |
| KOGA-S2  | 0.09                   | 0.93                              | 0.83                   | 0.91                              | 1.18                    |
| KOGA-S3  | 0.09                   | 0.92                              | 0.98                   | 0.91                              | 1.19                    |
| KOGA-S4  | 0.08                   | 0.93                              | 1.07                   | 0.92                              | 1.22                    |
| KOGA-W1  | 0.10                   | 0.92                              | 1.04                   | 0.93                              | 1.39                    |
| KOGA-W2  | 0.11                   | 0.91                              | 1.41                   | 0.91                              | 1.36                    |
| Sokcho   | 0.14                   | 0.89                              | 0.57                   | 0.94                              | 1.19                    |
| Average  | 0.09                   | 0.92                              | 0.86                   | 0.93                              | 1.23                    |

**Fig. 3.** Comparison of the averaged predicted wave spectra with the measurements.

which was split into the training and validation data. Here, the present network model was trained with only 70 % of the wave data, 174,600, while the other data was used to validate it. Beside of this, Table 1 shows that there are also the wave data for test, which were used to evaluate the model performance at other seas. Furthermore, the mini-batch was introduced to reduce the overfitting (Goodfellow et al., 2016), whose size is 300. The KL-divergence errors of the present neural network against the training and validation data are  $7.87 \times 10^{-2}$  and  $7.86 \times 10^{-2}$ , respectively. Then the present model was also applied to the test

data in order to see its feasibility at other seas. The accuracy of the present model was seen by estimating the KL-divergence error and Pearson's correlation coefficient of the predicted wave spectra for each wave measurement, and it was compared with those of JONSWAP wave spectrum (Table 3). Here, the peak enhancement factor,  $\gamma$  was determined by Hwang et al.'s (2021) formula. The  $\gamma$ -values for each wave measurement are listed in Table 3. This table shows that the values of  $\gamma$  are 1.15–1.39, which are comparable to those of Mazzaretto et al. (2022).

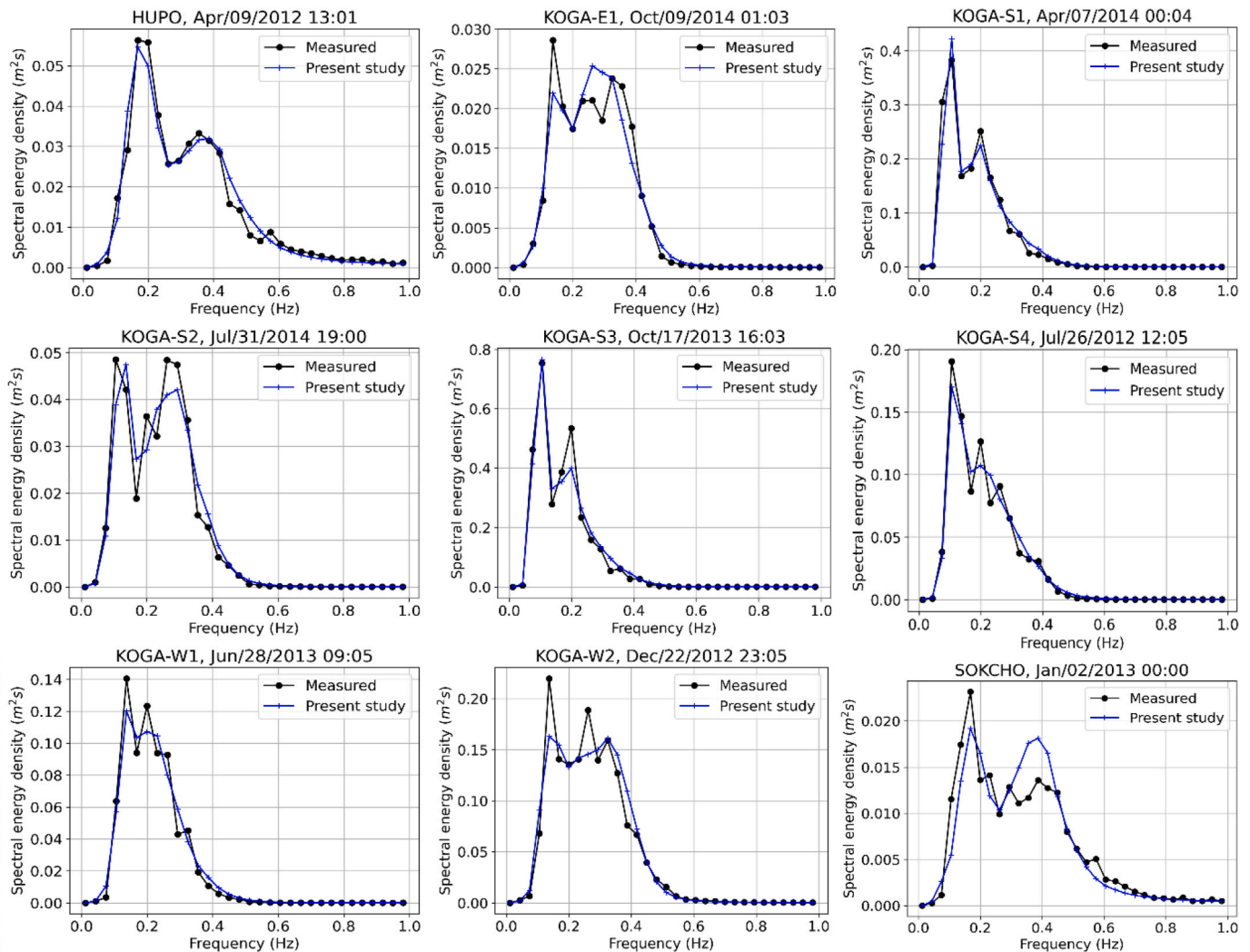


Fig. 4. Comparison of the predicted double-peaked wave spectra with measurements.

Averaging KL-divergence errors and Pearson's correlation coefficients for the training data, each average values are 0.08 and 0.94, while the corresponding ones for the test data are 0.1 and 0.92 (Table 3). This indicates that the present model is not limited to only the wave data considered in model training. Furthermore, Table 3 also shows that the average Pearson's correlation coefficient of the present model is a little smaller than that of JONSWAP wave spectrum. However, the KL-divergence error of JONSWAP wave spectrum is about 8.6 times larger than that of the present model. The spectral energy density over the frequencies lower than the peak frequency was underestimated. Then this made the KL-divergence error to be overestimated even though their contribution was small. Additionally, the present model aimed to minimize the KL-divergence error. Accordingly, the KL-divergence error of the present model is much smaller than the other. This indicates that the present model is comparable to JONSWAP wave spectrum, as also shown in Fig. 3.

On the other hand, the average wave spectra by the present model were compared with the corresponding of the measurement and JONSWAP wave spectrum (Fig. 3). Averaging wave spectra smoothed out the secondary peaks of wave spectra, and then only uni-modal spectra appear in Fig. 3. Despite that, the present model also provided the double peaked wave spectra (Fig. 4). This implies that the existence of secondary peaks in the wave spectra can be determined by wave parameters, because the present model predicts wave spectra from wave parameters. Similarly, Soares (1991) suggested the occurrence

Table 4

Accuracy of the predicted peak wave period.

| Locations | bias (s) | RMSE (s) | SI   | Pearson's correlation coefficient |
|-----------|----------|----------|------|-----------------------------------|
| Hupo      | -0.23    | 0.49     | 0.06 | 0.98                              |
| KOGA-E1   | -0.01    | 0.31     | 0.05 | 0.99                              |
| KOGA-S1   | 0.07     | 0.61     | 0.09 | 0.95                              |
| KOGA-S2   | -0.05    | 0.41     | 0.06 | 0.98                              |
| KOGA-S3   | -0.04    | 0.45     | 0.07 | 0.98                              |
| KOGA-S4   | 0.05     | 0.47     | 0.08 | 0.97                              |
| KOGA-W1   | 0.33     | 0.51     | 0.08 | 0.97                              |
| KOGA-W2   | 0.36     | 0.51     | 0.08 | 0.97                              |
| Sokcho    | -0.15    | 0.55     | 0.09 | 0.95                              |
| Average   | 0.04     | 0.48     | 0.07 | 0.97                              |

probability of the double peaked wave spectra based on the significant wave height. The relationship between the sea states and wave parameters will be introduced in the future study. Meanwhile, Sokcho is located in the JES, but the model accuracy at Sokcho is not higher than those of other seas (Table 3). This might result from that wave periods at Sokcho are 7–9 % shorter than those of the training data on average. These shorter wave periods lead to the larger wave steepness, extending the predicted wave spectra to the higher frequency (Fig. 3). At the same time the present model was trying to predict the peak frequency from maximum, significant, and mean waves, showing good agreements between the peak frequency of predicted wave spectra and the measured

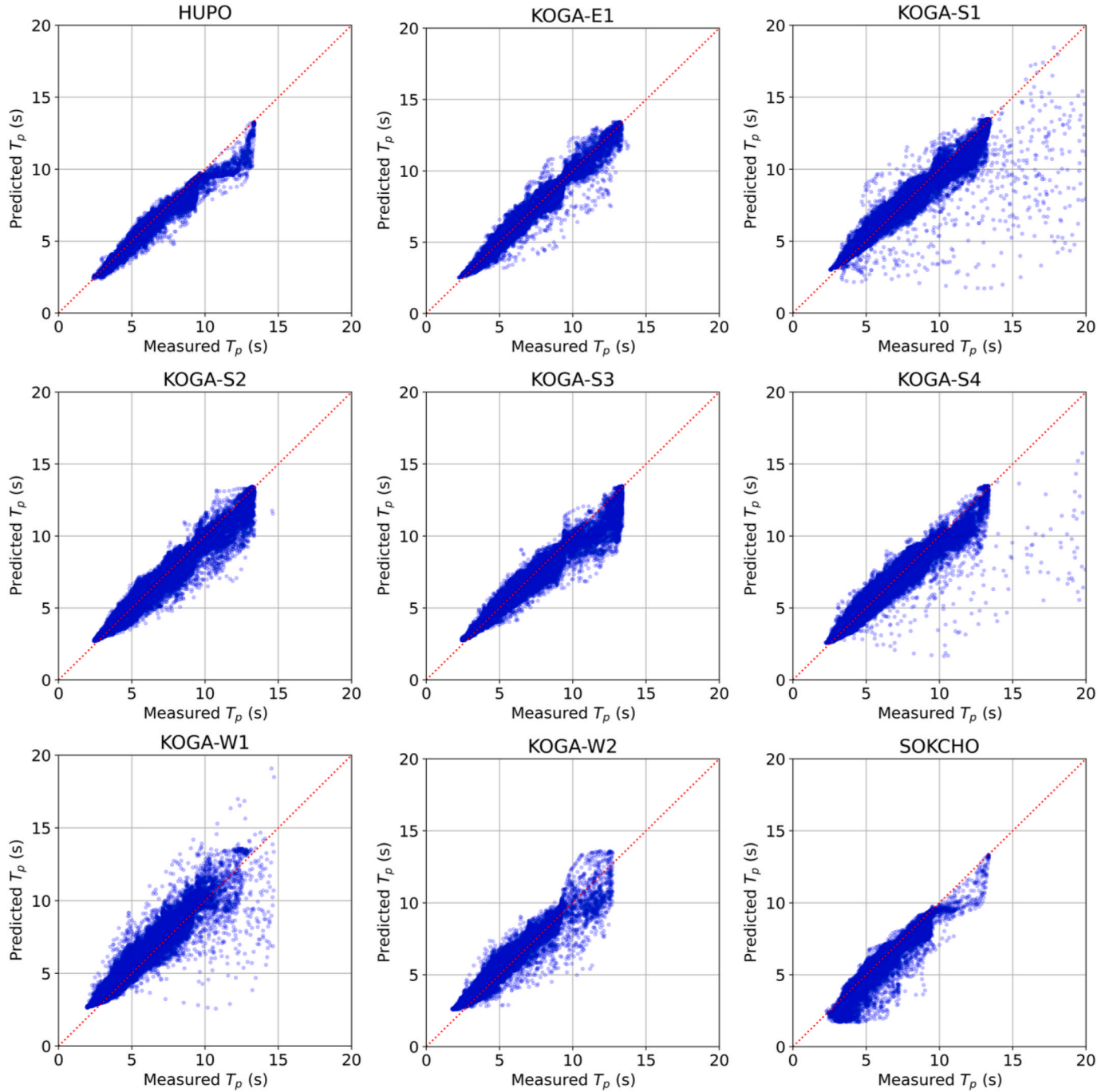


Fig. 5. Comparison of the predicted and measured  $T_p$ .

one. Additionally, the predicted peak wave period agrees with the measured one, even though it was underestimated (Table 4 and Fig. 5).

Besides, the predicted wave spectra were compared with those of JONSWAP wave spectrum and the measurements (Fig. 3). Fig. 3 shows that the wave spectra by the present model well agree with the measurements. However, the maximum values of wave spectra are somewhat inaccurate. Particularly, the present model underpredicted the spectral peak densities at KOGA-W1 and KOGA-W2. This can be explained by that the wave ages of waves at the Yellow Sea are younger than those of the JES owing to the shorter fetch length. Then the younger wave ages cause to increase the peakedness of wave spectrum (Young and Verhagen, 1996). Oppositely, the spectral peak densities at KOGA-S1, KOGA-S3, and KOGA-S4 were overestimated due to waves developed along with the larger fetch lengths. Although the wave data of

Hupo was included in model training, the shallow water depth also caused an overestimation in the spectral peak density at Hupo. Since the number of wave data is smaller than those of KOGA-E1 and KOGA-S2, the predicted spectral density at Hupo is larger than the measured one. In addition to this, the wave spectra by JONSWAP wave spectrum well predicted the spectral peak density regardless of the locations, because the spectral peak density was included in Hwang et al.'s (2021) formula on the peak enhancement factor.

### 2.3. Estimating peak wave period

As a peak wave period is one of the wave parameters from a wave spectrum, it was estimated by the present model, and then compared with the measured one. The accuracy of the predicted peak period is

**Table 5**

Comparison of the spectral shape parameters of wave spectra predicted by the present model with the measured ones.

| Locations | Spectral shape parameter | Bias  | RMSE | SI   | Pearson's correlation coefficient |
|-----------|--------------------------|-------|------|------|-----------------------------------|
| Hupo      | $Q_p$                    | -0.17 | 0.39 | 0.18 | 0.66                              |
|           | $\varepsilon$            | -0.06 | 0.07 | 0.04 | 0.90                              |
| KOGA-E1   | $Q_p$                    | -0.22 | 0.42 | 0.17 | 0.65                              |
|           | $\varepsilon$            | 0.01  | 0.04 | 0.05 | 0.93                              |
| KOGA-S1   | $Q_p$                    | -0.03 | 0.33 | 0.17 | 0.69                              |
|           | $\varepsilon$            | 0.02  | 0.04 | 0.05 | 0.89                              |
| KOGA-S2   | $Q_p$                    | -0.14 | 0.36 | 0.17 | 0.73                              |
|           | $\varepsilon$            | 0.02  | 0.04 | 0.05 | 0.94                              |
| KOGA-S3   | $Q_p$                    | -0.11 | 0.35 | 0.17 | 0.73                              |
|           | $\varepsilon$            | 0.02  | 0.04 | 0.05 | 0.93                              |
| KOGA-S4   | $Q_p$                    | -0.10 | 0.37 | 0.18 | 0.71                              |
|           | $\varepsilon$            | 0.02  | 0.04 | 0.05 | 0.92                              |
| KOGA-W1   | $Q_p$                    | -0.39 | 0.58 | 0.18 | 0.74                              |
|           | $\varepsilon$            | 0.02  | 0.05 | 0.07 | 0.89                              |
| KOGA-W2   | $Q_p$                    | -0.41 | 0.59 | 0.20 | 0.76                              |
|           | $\varepsilon$            | 0.03  | 0.06 | 0.07 | 0.87                              |
| Sokcho    | $Q_p$                    | -0.40 | 0.59 | 0.22 | 0.52                              |
|           | $\varepsilon$            | -0.04 | 0.06 | 0.05 | 0.80                              |
| Average   | $Q_p$                    | -0.22 | 0.44 | 0.18 | 0.69                              |
|           | $\varepsilon$            | 0.00  | 0.05 | 0.05 | 0.90                              |

presented in terms of bias, RMSE (Root Mean Square Error), SI (Scatter Index), and Pearson's correlation coefficient (Table 4). Here, the bias is defined as an average value of subtraction of the measurements from the predictions. According to Table 4, the RMSEs of the predicted peak wave periods range from 0.31 s to 0.61 s. Its average value is 0.48 s, and the corresponding Pearson's correlation coefficient is 0.97. This indicates that the predicted peak wave periods show good agreements with the measured ones.

On the other hand, the peak periods of KOGA-W1 and KOGA-W2 were overpredicted. This came from the differences in the wave characteristics between the JES and other seas, particularly, the fetch length, as like the underprediction in the spectral peak density. Beside of this, the RMSE of KOGA-S1 is much larger than the average RMSE (Table 4). Owing to the geographical characteristics of KOGA-S1, southerly long waves were developed along with the long fetch and this fetch length is much longer than those of the wave measurements used in model training. Consequently, the present model could not predict the peak wave periods of several long waves (Fig. 5). Furthermore, Fig. 5 shows the low accuracy of peak wave periods larger than 10 s, because the frequency resolution of the present model was not fine enough to present the wave periods precisely.

### 3. Discussion

Meanwhile, the spectral peakedness parameter,  $Q_p$  and the spectral bandwidth parameter,  $\varepsilon$  were estimated to compare the spectral shapes of the predicted wave spectra with the measurements (Table 5). Here, Goda (1970) suggested  $Q_p = (2/m_0^2) \int_0^\infty f S^2(f) df$  using the spectral energy density and the zeroth order moment of wave spectrum,  $m_0$ , where  $n$ -th order wave spectrum is given as  $m_n = \int_0^\infty f^n S(f) df$ .  $Q_p$  measures the peakedness of wave spectrum, which is strongly correlated with the peak enhancement factor of JONSWAP wave spectrum (Ahn et al., 2013; Hwang et al., 2021). Besides, the form of  $\varepsilon$  was given as  $\varepsilon = \sqrt{1 - m_2^2/(m_0 m_4)}$  (Cartwright and Longuet-Higgins, 1956). This term is used to measure the energy spreading of a wave spectrum, being broadly spread for a wave spectrum with the larger value of  $\varepsilon$ . The  $\varepsilon$ -value for ocean waves lies in the range between 0.4 and 0.8 (Ochi, 1998).

In the present study,  $Q_p$  and  $\varepsilon$  of the predicted and measured wave spectra for each wave measurement were compared to each other, and then the accuracy of these parameters was given in Table 5. According to Table 5, the Pearson's correlation coefficients of  $Q_p$  are 0.52–0.76,

resulting from that the activation function at the last layer of present model, softmax has softened wave spectra (Charniak, 2018). Besides, the present model underpredicted  $Q_p$  in overall (Table 5). In particular, the biases of  $Q_p$  at KOGA-W1 and KOGA-W2 are lower than the average, -0.22, as also shown in Fig. 6. This is consistent with the underestimation in the spectral peak density (Fig. 3). On the other hand, the spectral peak density by the present model at Sokcho is similar to the measured one (Fig. 3), while the present model underestimated  $Q_p$ . According to Table 5, the Pearson's correlation coefficient of  $Q_p$  at Sokcho is much smaller than those of other sites. This was caused by the underestimation in the term,  $\int_0^\infty f S^2(f) df$ , because the term was more spread around the peak frequency.

On the other hand, the sharper wave spectra seem to be narrowly spread, but  $Q_p$  and  $\varepsilon$  are independent to each other (Ahn et al., 2013). Thus the Pearson's correlation coefficients of  $\varepsilon$  are 0.80–0.94 unlike  $Q_p$  (Table 5). The strong correlation between the measured and predicted spectral bandwidths can be also seen in Fig. 7. This indicates that the present model can predict the overall shape of wave spectra well (Fig. 3), even though the present model somewhat underpredicted the spectral bandwidth at Sokcho. In addition to this, the Pearson's correlation coefficients of  $\varepsilon$  for the Yellow Sea and the South Sea are not smaller than those of other data like the case of JONSWAP wave spectrum (Table 6). This implies that the present model is not limited to the local sea.

Furthermore, the spectral shape parameters of wave spectra by the JONSWAP wave spectrum were also compared with the measured ones (Table 6). According to Table 6, the Pearson's correlation coefficients of  $Q_p$ , 0.71–0.86, are much higher than those of the present model. Its average value is 0.78, which is 13.0 % larger than that of the present model, because Hwang et al.'s (2021) formula contributed to improve the spectral shape of JONSWAP wave spectrum. On the other hand, the average RMSE of JONSWAP, 0.39, is a little larger than that of the present model, 0.44. This indicates that the errors of  $Q_p$  by both models are similar to each other, but  $Q_p$  values by the present model are more scattered around the measurements (Fig. 6). The present model can predict  $Q_p$  less than 2, while  $Q_p$  values of JONSWAP wave spectrum converge to a value around 2 due to the frequency resolution (Fig. 6). Beside of this, the  $\varepsilon$ -values of the JONSWAP wave spectrum model, 0.82–0.90, are a little smaller than those of the present model. Particularly, its average value, 0.87 is 3.3 % smaller than that of the present model, 0.9. However, the  $\varepsilon$ -values converge to a value around 0.8 due to the frequency resolution.

Summarizing this, the present model poorly estimated  $Q_p$ , even though the  $\varepsilon$ -values of the predicted wave spectra are a bit more accurate than that of JONSWAP wave spectrum. From this, it can be seen that the present model is not better than JONSWAP wave spectrum. However, the present model does not require the spectral wave parameters such as the spectral peak density to predict wave spectra, while the spectral wave parameters was incorporated into JONSWAP wave spectrum model for better suitability. Additionally, the present model can reproduce the double peaked wave spectra, despite not assuming the spectral form. At this point, the present model is more advantageous than JONSWAP wave spectrum. Furthermore, the present model can estimate the wave peak period from the wave parameters by zero-crossing analysis throughout the wave spectra by the present model.

### 4. APPLICATION

Although widely applied to the coastal engineering, the wave spectrum is particularly used as an incident wave condition of numerical and hydraulic experiments. For this reason, the wave spectra by the present model were employed to the wave hindcasting in Young-il bay to see its applicability. Here, the study area is presented in Fig. 8. At the entrance of Young-il bay, the wave observation had been performed from 1991 to 2004, whose location is presented as W1 in Fig. 8. During the measurement, its maximum significant wave height, 5.74 m was reached on

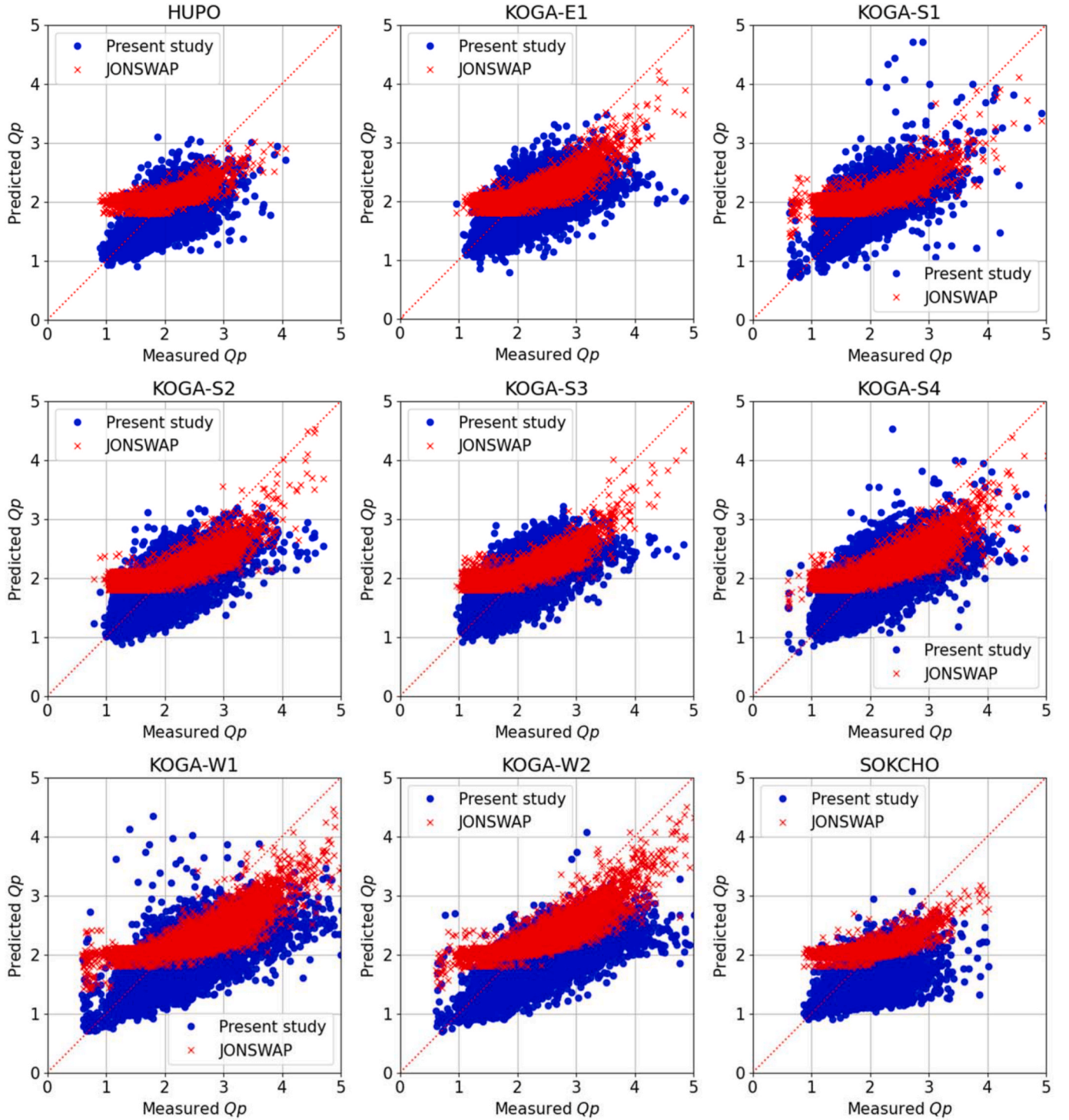


Fig. 6. Comparison of the predicted and measured  $Q_p$ .

19th September 1999. Accordingly, the present numerical simulation has been carried out for the period from 15th September 1999 to 30th September 1999.

In the present study, [Chun and Ahn's \(2017\)](#) model has been implemented. Since the numerical model was extended to shallow water, it could be applied to the wave prediction in coastal areas. Its governing equation is as follows:

$$\frac{\partial F}{\partial t} + \frac{\partial(c_\lambda F)}{\partial \lambda} + \frac{1}{\cos \varphi} \frac{\partial(\cos \varphi c_\varphi F)}{\partial \varphi} + \omega_r \frac{\partial(c_{\omega_r} F)}{\partial \omega_r} + \frac{\partial(c_\theta F)}{\partial \theta} = S \quad (11)$$

where  $F(\lambda, \varphi, \omega_r, \theta)$  is the wave energy density,  $\lambda$  and  $\varphi$  denote the longitude and latitude, respectively, and  $\omega_r$  is the relative frequency, which is given as  $\omega_r^2 = g\tilde{k}\tanh\tilde{k}h$  with the wave number,  $\tilde{k}$ , and the local water depth,  $h$ .  $\theta$  is the wave direction measured clockwise from the north.  $c_\lambda$ ,  $c_\varphi$ ,  $c_{\omega_r}$ , and  $c_\theta$  are the group velocities in each space, respectively. In Eq. (10),  $S$  represents the wave generation and dissipation like the following:

$$S = S_{in} + S_{ds} + S_{nl3} + S_{nl4} + S_{br} + S_{bf} \quad (12)$$

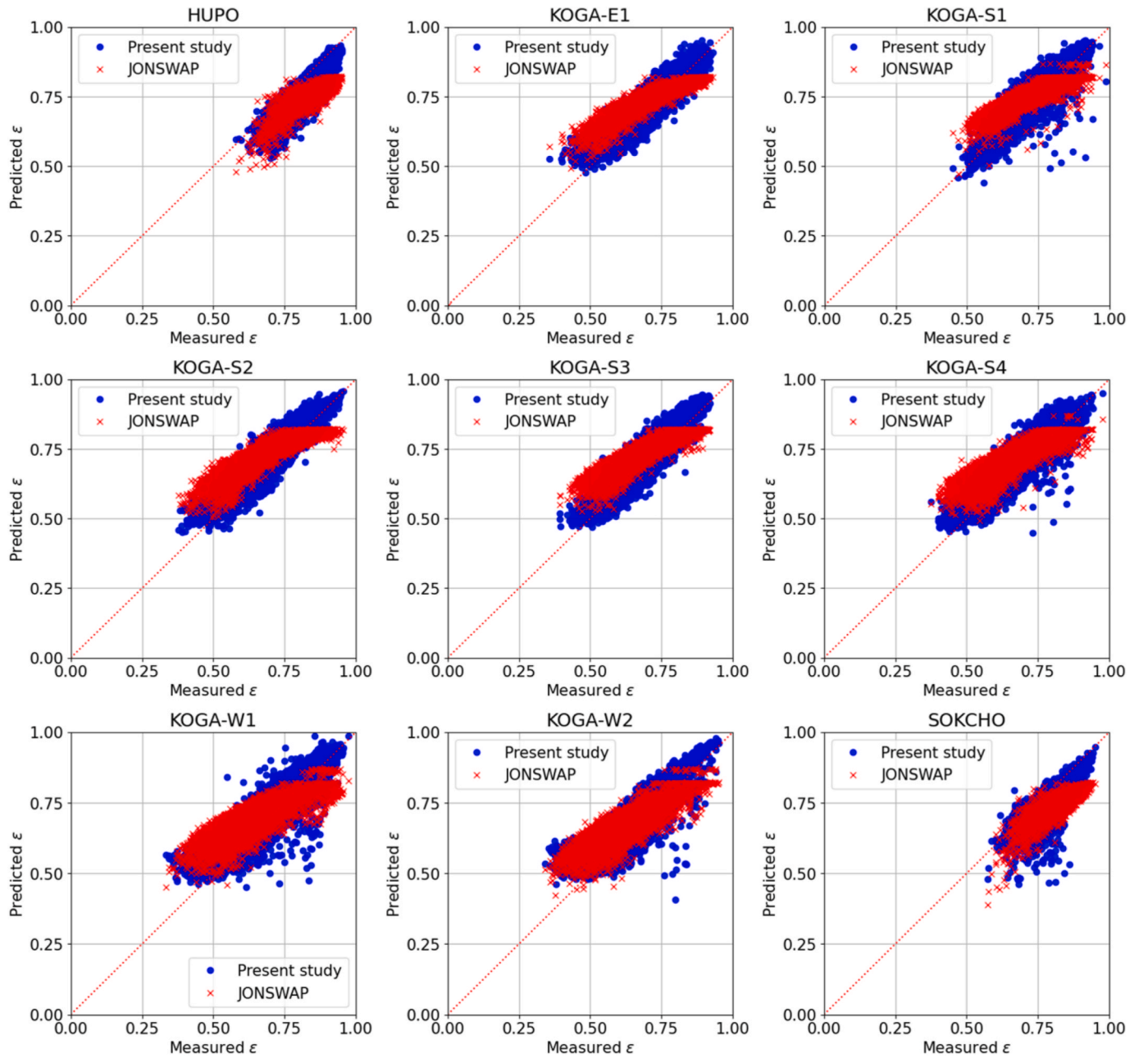


Fig. 7. Comparison of the predicted and measured  $\varepsilon$ .

where  $S_{in}$  is the wave generation by the wind,  $S_{ds}$  is the white capping dissipation,  $S_{nl3}$  and  $S_{nl4}$  denote the triad and quadruplet wave-wave interaction terms, respectively,  $S_{br}$  is the wave breaking dissipation, and  $S_{bf}$  is the bottom friction dissipation. The descriptions of these terms were given in [Chun and Ahn \(2017\)](#).

The computation domain of the present numerical simulation is presented in [Fig. 8](#), where the water depth was from the nautical chart by KHOA (Korea Hydrographic and Oceanographic Agency). The spatial grid size was 100 m. A wave spectrum was divided into 12 frequencies and 26 directional bins. Specifically, the frequency increases in the logarithmic scale, and the direction of the wave spectrum ranged from  $-90^\circ$  to  $90^\circ$  with an increment of  $7.2^\circ$ . The time increment was 60 s. The wind data measured by KMA (Korea Meteorological Administration) was used to the present wave hindcasting.

The incident wave of the numerical simulation was obtained by applying the present model to the wave measurement. Additionally, the wave spectra by JONSWAP wave spectrum were also considered as other

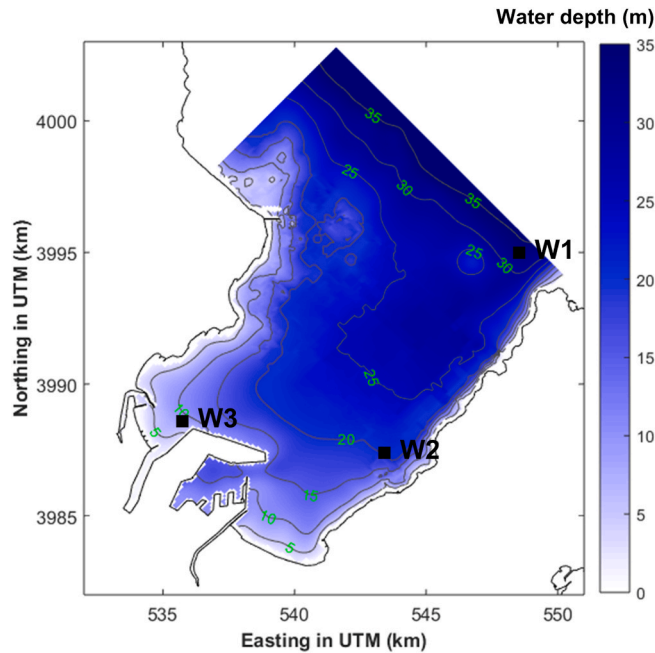
incident wave condition, and then numerical experiments for two wave conditions have been carried out. The comparison of the present neural network and JONSWAP wave spectrum was given in [Fig. 9](#). This figure shows that the wave spectrum from the present neural network is sharper than that of JONSWAP wave spectrum. Here, the peak enhancement factor of JONSWAP wave spectrum was 1.19, the average value of peak enhancement factor in the JES ([Table 3](#)). In addition to this, [Mitsuyasu et al.'s \(1975\)](#) directional spreading function was employed to two incident waves of the present numerical simulation.

After the numerical simulation, the predicted significant wave height and period at W2 and W3 were compared with the measurements, whose locations are presented in [Fig. 8](#). Here, the predicted significant wave period was estimated by [Chun and Suh's \(2018\)](#) formula,  $T_s = 0.76T_{-1.0}^{1.11}$ , where  $T_{-1.0}$  is expressed as  $T_{-1.0} = m_{-1}/m_0$ . [Fig. 10](#) shows that there are overall agreements between the numerical results and the measurements. In particular, the significant wave height and period at the peak of storm are well predicted. Besides, the predicted significant

**Table 6**

Comparison of the spectral shape parameters of wave spectra predicted by JONSWAP wave spectrum with the measured ones.

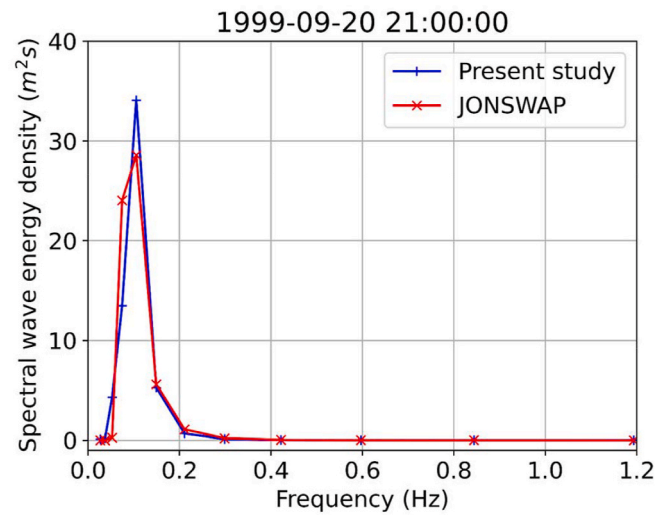
| Locations | Spectral shape parameter | Bias  | RMSE | SI   | Pearson's correlation coefficient |
|-----------|--------------------------|-------|------|------|-----------------------------------|
| Hupo      | $Q_p$                    | 0.15  | 0.40 | 0.19 | 0.71                              |
|           | $\varepsilon$            | -0.08 | 0.09 | 0.04 | 0.83                              |
| KOGA-E1   | $Q_p$                    | -0.03 | 0.33 | 0.15 | 0.81                              |
|           | $\varepsilon$            | 0.04  | 0.06 | 0.07 | 0.91                              |
| KOGA-S1   | $Q_p$                    | 0.13  | 0.37 | 0.18 | 0.75                              |
|           | $\varepsilon$            | 0.03  | 0.06 | 0.06 | 0.87                              |
| KOGA-S2   | $Q_p$                    | 0.12  | 0.38 | 0.18 | 0.79                              |
|           | $\varepsilon$            | 0.05  | 0.07 | 0.07 | 0.90                              |
| KOGA-S3   | $Q_p$                    | 0.13  | 0.39 | 0.19 | 0.79                              |
|           | $\varepsilon$            | 0.06  | 0.07 | 0.07 | 0.89                              |
| KOGA-S4   | $Q_p$                    | 0.10  | 0.38 | 0.18 | 0.81                              |
|           | $\varepsilon$            | 0.05  | 0.07 | 0.07 | 0.88                              |
| KOGA-W1   | $Q_p$                    | -0.11 | 0.45 | 0.19 | 0.83                              |
|           | $\varepsilon$            | 0.06  | 0.08 | 0.09 | 0.84                              |
| KOGA-W2   | $Q_p$                    | 0.03  | 0.42 | 0.19 | 0.86                              |
|           | $\varepsilon$            | 0.05  | 0.07 | 0.08 | 0.85                              |
| Sokcho    | $Q_p$                    | 0.12  | 0.42 | 0.20 | 0.71                              |
|           | $\varepsilon$            | -0.07 | 0.08 | 0.04 | 0.82                              |
| Average   | $Q_p$                    | 0.07  | 0.39 | 0.18 | 0.78                              |
|           | $\varepsilon$            | 0.02  | 0.07 | 0.07 | 0.87                              |



**Fig. 8.** Computational domain of the wave hindcasting for Young-il bay and locations of the wave measurements.

wave height is more accurate than that of the significant wave period, because the significant wave period was estimated from the numerical results after implementing the numerical model, unlike the significant wave height. On the other hand, the significant wave heights from both numerical models are similar to each other, even though the significant wave periods of two cases are different. This results from that the spectral shape affects to the significant wave period by two terms,  $m_{-1}$  and  $m_0$ . This is shown in Table 7 and Fig. 11.

According to Table 7, the accuracies of both numerical models for the significant wave heights are nearly same, as shown in Fig. 10. In particular, the RMSEs of significant wave heights from both numerical models are the same, while the perturbations in the predicted significant wave heights by JONSWAP wave spectrum are a little larger than the other, as shown in Fig. 10. On the other hand, the RMSE of significant



**Fig. 9.** Comparison of wave spectra from the present neural network and JONSWAP at the time when the significant wave height was observed.

wave period by the present neural network is 7.08 % smaller than the other, and its Pearson's correlation coefficient is 6.33 % larger on average (Table 7).

## 5. Conclusion

Owing to the importance of wave spectra, several researchers had proposed methods to estimate wave spectra from the wave parameters (Hasselmann et al., 1973; Hwang et al., 2021; Naithani and Deo, 2005; Namecar and Deo, 2006). Among them, JONSWAP wave spectrum is widely applied to predict wave spectra, and its suitability can be improved by modifying parameters (Mazzaretto et al., 2022). On the other hand, Hwang et al. (2021) estimated wave spectra using the individual wave parameters from the zero-crossing analysis. Since the specific form of wave spectrum was not assumed in Hwang et al. (2021), various spectral forms can be reproduced by this method. However, it is somewhat inefficient, requiring many wave parameters.

For this reason, DeCNN, one of neural network methods, was employed to predict wave spectra from wave parameters by the zero-crossing analysis, maximum, significant, and average waves, in the present study. Previously, the neural network method had been also applied to predict wave spectra (Naithani and Deo, 2005; Namecar and Deo, 2006), but their methods are different from that of the present model, not requiring the wave parameters by the spectral analysis. Additionally, the number of data is larger than that of previous studies. It is common that a large amount of data is helpful to improve the accuracy and generality of neural network. The present neural network was trained with the wave measurements in the JES, Hupo, KOGA-E1, and KOGA-S2. Then the model was applied to the wave data in the Yellow Sea and the South Sea to see its feasibility at other seas. Even though the wave characteristics of these seas are different from that of the JES, there were also good agreements between the predicted and measured wave spectra. This shows the generality of the present model, particularly, considering that the number of the test data is larger than that of the train data. However, the present model should be validated against the wave data at other seas, because of the limit in the data-driven approach. Additionally, the overall spectral shapes by the present method well agree with the measured ones, even though the peakedness parameters of wave spectra by the present method were underpredicted. This might result from that the activation function used in the last layer of present method, softmax, has softened the spectral shape.

Besides, the present method was compared with JONSWAP wave

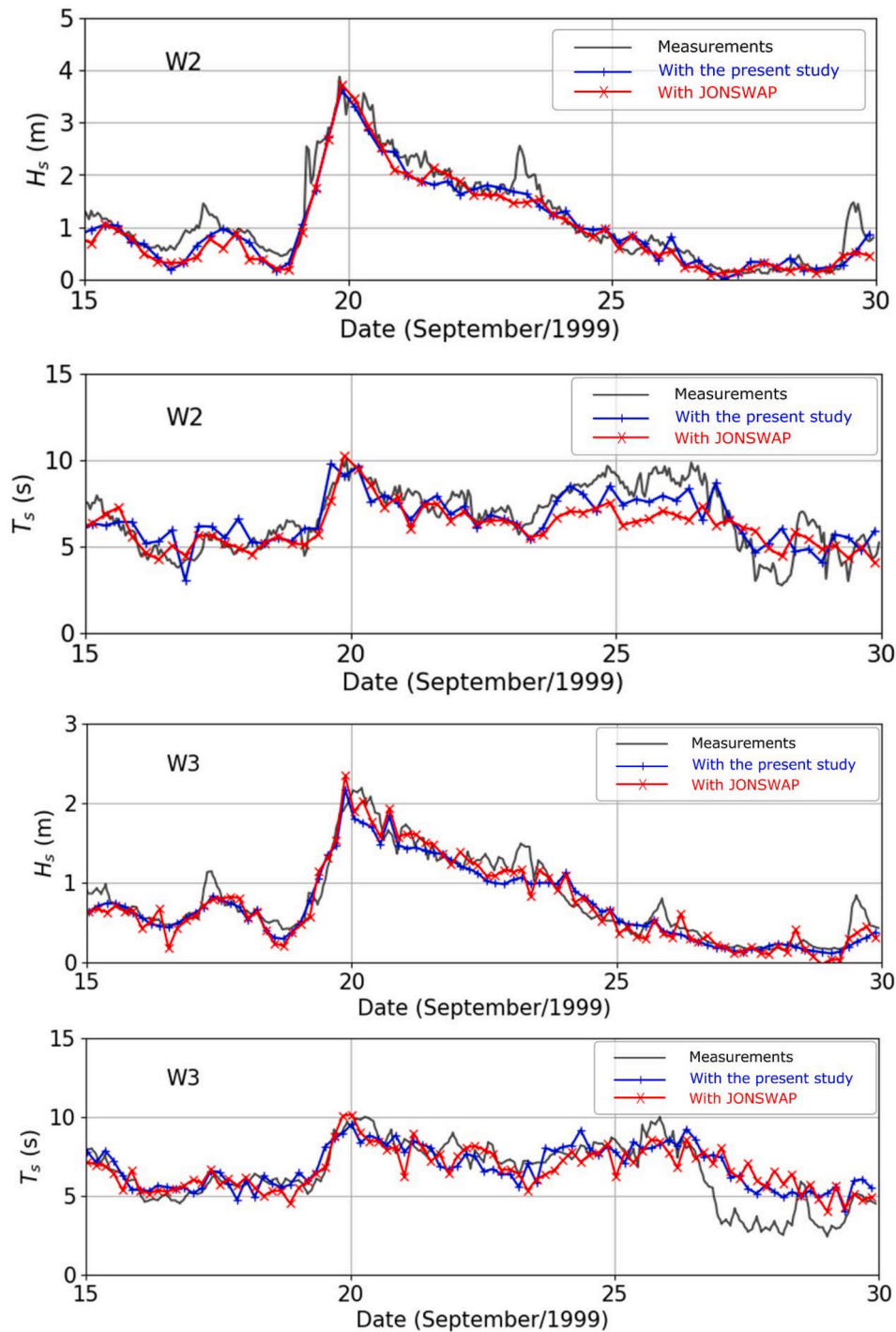


Fig. 10. Comparison of the predicted significant wave height and period with the measurements.

spectrum, where the peak enhancement factor of JONSWAP wave spectrum was evaluated by Hwang et al.'s (2021) formula. Since the spectral peak density was incorporated into Hwang et al.'s (2021) formula, the peakedness parameter of wave spectra by JONSWAP wave spectrum was more accurate than that of the present model. In addition to this, the accuracy of spectral bandwidth parameter by JONSWAP wave spectrum was a bit lower than that of the present model. This might be the limitation of the present model, if considering the complexity of the present model. However, the local wave characteristics and other wave parameters are not required for better suitability, when the present model is applied to other seas. Furthermore, wave parameters from the spectral analysis were not required to build the

present model. The present model can produce the double peaked wave spectra, not assuming the spectral form. From this, the applicability of the present model might not be worse than that of JONSWAP wave spectrum.

For this reason, the present method was employed to the wave hindcasting in Young-il bay to see its applicability. In the present numerical simulation, the incident waves were given by the present method and JONSWAP wave spectrum. According to the present study, the calculated significant wave heights for both cases well agree with the measurements. Additionally, significant wave periods were estimated from the numerical results using Chun and Suh's (2018) formula. Then the predicted significant wave periods from both cases were compared

**Table 7**  
Accuracy of wave hindcasting in Young-il bay.

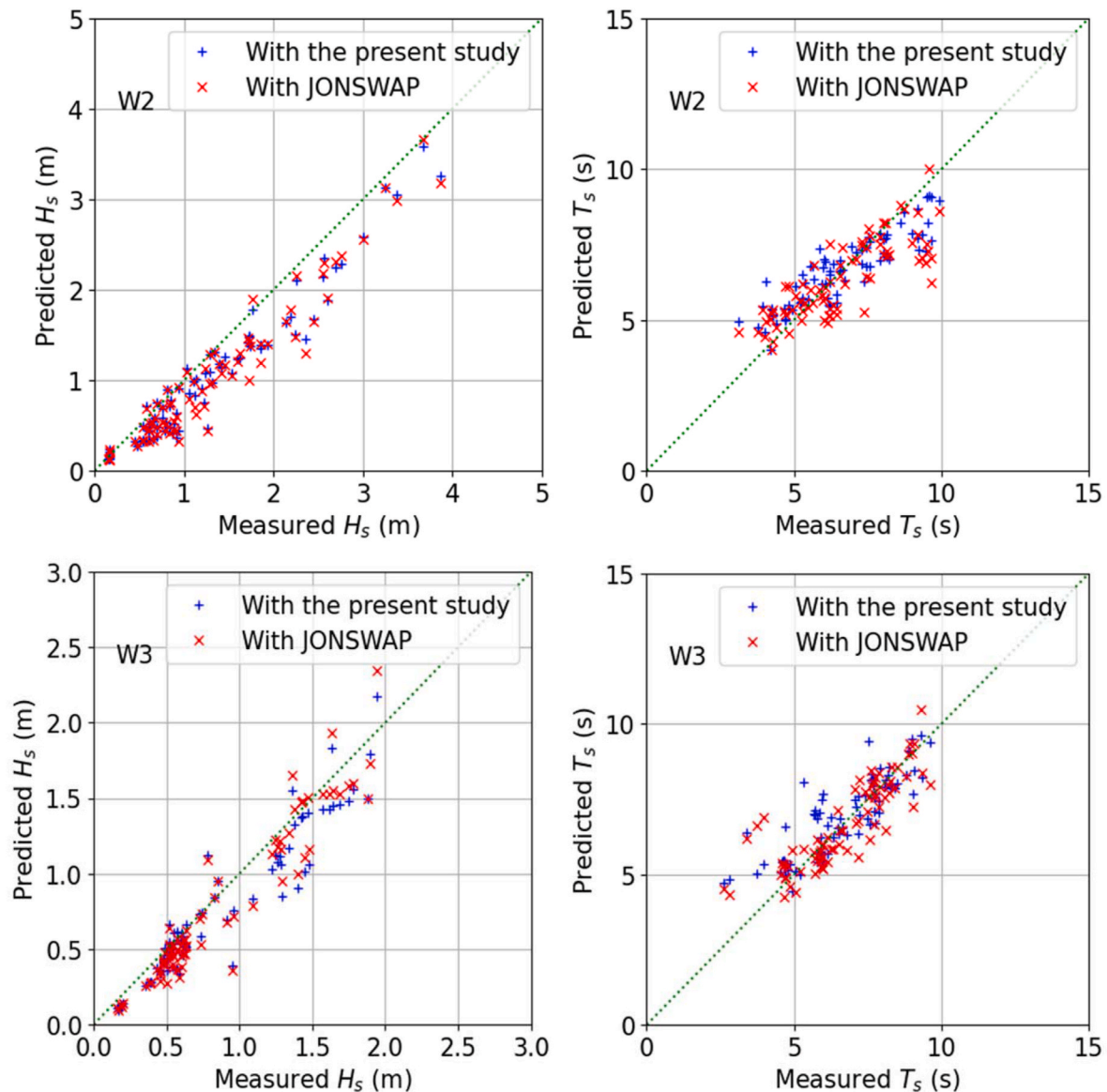
| Locations | Incident wave condition | Physical quantities | bias    | RMSE   | SI   | Pearson's correlation coefficient |
|-----------|-------------------------|---------------------|---------|--------|------|-----------------------------------|
| W2        | Present model           | $H_s$               | -0.26 m | 0.34 m | 0.16 | 0.97                              |
|           |                         | $T_s$               | -0.23 s | 1.04 s | 0.15 | 0.85                              |
|           | JONSWAP                 | $H_s$               | -0.28 m | 0.37 m | 0.18 | 0.96                              |
|           |                         | $T_s$               | -0.28 s | 1.14 s | 0.17 | 0.78                              |
| W3        | Present model           | $H_s$               | -0.12 m | 0.21 m | 0.20 | 0.94                              |
|           |                         | $T_s$               | -0.07 s | 0.93 s | 0.14 | 0.83                              |
|           | JONSWAP                 | $H_s$               | -0.14 m | 0.23 m | 0.21 | 0.93                              |
|           |                         | $T_s$               | -0.05 s | 0.98 s | 0.15 | 0.80                              |

with the measurements. The significant wave period of the case with the present method was more accurate than the other, owing to the more accurate spectral shape of case with the present method.

On the other hand, the directional distribution of incident waves was not given by the present neural network model. Instead, the bell-shaped parameter model, a symmetric directional spreading function, was applied to the present numerical experiment, which is common in the spectral wave model (The SWAN team, 2024). However, the waves with asymmetric directional spreading were also observed in the coastal region (Young, 2010). Lee et al. (2010) have proposed a parameter model for the asymmetric directional spreading function, but it is hard to find its application to other field measurements. In the future, the present neural network model will be extended to the directional wave spectrum, and compared with this asymmetric directional spreading function. Beside of this, it will be carried out to classify the sea states by the relationship among wave parameters.

#### CRediT authorship contribution statement

**Soonmi Hwang:** Writing – review & editing, Writing – original draft, Validation, Investigation, Formal analysis. **Jung Lyul Lee:** Writing –



**Fig. 11.** Scatter diagram of predicted versus measured significant wave height and period at W2 and W3.

review & editing, Writing – original draft, Supervision. **Hwusub Chun:** Writing – review & editing, Writing – original draft, Validation, Methodology.

### Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

### Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.oceaneng.2024.118331>.

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